TECAN.

Command Set

Pick & Place Arm

P & P

Corresponds to Firmware Version: V1.04

Doc. No: 391412
V1.04
3 October, 2000

Contents:

1 REVISION HISTORY 4

2 INTRODUCTION 5

3 DEFINITION OF TERMS 5

4 COMMANDS 5

4.1 POSITIONING, MOVE AND ACTION COMMANDS: 5

4.1.1 PIF 5
4.1.2 PIA 5
4.1.3 PIX InitSpeed 5*
4.1.4 PIY InitSpeed 6*
4.1.5 PIZ InitSpeed 6*
4.1.6 PIR 6
4.1.7 PAX X 6*
4.1.8 PAY Y 7*
4.1.9 PAZ Z 7*
4.1.10 PAR R 7*
4.1.11 PAG G 7*
4.1.12 PRX +/- X 8*
4.1.13 PRY +/- Y 8*
4.1.14 PRZ +/- Z 8*
4.1.15 PRR +/- R 8*
4.1.16 PRG +/- G 8*
4.1.17 MAX X, SlowSpeed* 9*
4.1.18 MAY Y, SlowSpeed* 9*
4.1.19 MAZ Z, SlowSpeed* 9*
4.1.20 MAR R, SlowSpeed* 10*
4.1.21 MAG G, SlowSpeed* 10*
4.1.22 MRX +/- X, SlowSpeed* 10*
4.1.23 MRY +/- Y, SlowSpeed* 10*
4.1.24 MRZ +/- Z, SlowSpeed* 11*
4.1.25 MRR +/- R 11*
4.1.26 MRG +/- G, ... 11*
4.1.27 PAA X,Y*,Z*,R*,G* 12*
4.1.28 PRA X,Y*,Z*,R*,G* 12*
4.1.29 MAA X,Y*,Z*,R*,G* 12*
4.1.30 MRA X,Y*,Z*,R*,G* 13*
4.1.31 AGR GripSearch, GripOpen* 13*
4.1.32 APC GripSearch,GripOpen*,Zpick*,Zup*,Rotation*, RotComp* 14*
4.1.33 APL GripOpen, Zplace*, Zup*, Rotation*, TubCheck*, WipeOffDist*, TubeLeftChck* 14*
4.1.34 ATD Diameter 15
4.1.35 ARX 15
4.1.36 ARY 15
4.1.37 ARZ 15

4.2 SET COMMANDS 16

4.2.1 SGP PWMLimit, GripSpeed*, MinSpace*, MaxSpace* 16*
4.2.2 SFX EndSp, Accel* 16*
4.2.3 SFY EndSp, Accel* 16*
4.2.4 SFZ EndSp, Accel* 16*
4.2.5 SFR EndSp, Accel* 17*
4.2.6 SOX InitOffset 17
4.2.7 SOY InitOffset 17
4.2.8 SOZ InitOffset 17

4.2.9 SOR InitOffset ... *18*
4.2.10 SAX ScaleAdjust ... *18*
4.2.11 SAY ScaleAdjust ... *18*
4.2.12 SXD Displacement ... *18*
4.2.13 SMX Range ... *19*
4.2.14 SMY Range ... *19*
4.2.15 SMZ Range ... *19*
*4.2.16 SCO AxisNo** ... *20*
4.2.17 SCS SubmergeRange ... *20*
4.2.18 SSO OnOff ... *20*
4.3 REPORT COMMANDS ... 21
4.3.1 RFV Selector ... *21*
4.3.2 RDF Selector ... *21*
4.3.3 REE ... *21*
*4.3.4 RPX Selector** ... *22*
*4.3.5 RPY Selector** ... *22*
*4.3.6 RPZ Selector** ... *23*
*4.3.7 RPR Selector** ... *23*
*4.3.8 RPG Selector** ... *24*
4.3.9 RGP Selector ... *24*
4.3.10 RTD ... *25*
4.3.11 RXD ... *25*
5 ERROR CODES AND MESSAGES ... 26

1 Revision History

D.Bischof	Changes in gate part (see C&G)	PPx_V1.01		
D.Bischof	diameter calculation improved	PPx_V1.03		
	Z-accelleration reduced by 25%			
	Z-speed reduced by 25%			
D.Bischof	New features in <APL> command	PPx_V1.04		
	with WipeOff andTubeLeftcheck			

2 Introduction

For Commands regarding General Gates see Command Set “General Gate Command Structure” (New Conveyor), CS_Gate_V1.0.

3 Definition of Terms

4 Commands

4.1 POSITIONING, MOVE AND ACTION COMMANDS:

4.1.1 PIF

FAKE INITIALIZATION X/Y/Z/R/G - AXIS:

The PIF command initializes all axes. However, no physical initialization is executed. The current position of each axis are still valid. If power is disconnected and turned on, using the *PIF* command the axis positions are not defined! CAUTION: Incorrect usage of this command may damage the instrument.

EXAMPLE: #W2PIF

Fake initialization of the P&P.

RESPONSE: none

GENERATED ERRORS: none

4.1.2 PIA

POSITION INITIALIZATION X/Y/Z/R/G - AXES:

This command initializes the P&P X/Y/Z/Rotator and Gripper axis. No movement command for the arm will be accepted before the *PIA* command is executed. When the *PIA* command is entered, all axis move to their initialization positions with a default init speed. After the init position of an axis has been detected, it moves the defined offset steps in the opposite direction.

EXAMPLE: #W2PIA

Initialization of all arm axis.

4.1.3 PIX InitSpeed*

POSITION INITIALIZATION X - AXIS:

This function re-initializes only the X-axis. An override motor speed for axis initialization may be placed at the end of the command. A command without any parameter uses the default speed to initialize.

InitSpeed: speed in 0.1 mm/s [5...1500] (default = 205)

EXAMPLE: #W2PIX250

Initialization of the P&P X-axis with an init speed of 25 mm/s.

RESPONSE: none

GENERATED ERRORS: (1) init failed, (3) invalid operand, (5) device not implemented

4.1.4 PIY InitSpeed*

POSITION INITIALIZATION Y - AXIS:

This function reinitializes only the Y-axis on the P&P. An override motor speed for axis initialization may be placed at the end of the command. A command without any parameter uses the default speed for initialization.

InitSpeed: speed in 0.1 mm/s [5...1500] (default = 505)

EXAMPLE: #W2PIY100

Initialization of the P&P Y- axis with an init speed of 10 mm/s.

RESPONSE: none

GENERATED ERRORS: (1) init failed, (3) invalid operand, (5) device not implemented

4.1.5 PIZ InitSpeed*

POSITION INITIALIZATION Z - AXIS:

This function reinitializes only the Z-axis on the P&P. An override motor speed for axis initialization may be placed at the end of the command. A command without any parameters uses the default speed and all tips to initialize.

InitSpeed: speed in 0.1 mm/s [5...1500] (default = 296)

EXAMPLE: #W2PIZ170

Initialization of the P&P Z-axis with an init speed of 17 mm/s.

RESPONSE: none

GENERATED ERRORS: (1) init failed, (3) invalid operand, (5) device not implemented

4.1.6 PIR

POSITION INITIALIZATION ALL DEPENDENT AXES:

This function initializes all dependent axis of the P&P. First the Z-axis moves up, then the Y-axis moves to a save position, the rotator and the gripper are initialized afterwards. Finally the Z-axis is initialized again and moves to it's offset position. It uses the default speed for initialization.

EXAMPLE: #W2PIR

Initialization of the P&P Z-axis, rotator and gripper.

RESPONSE: none

GENERATED ERRORS: (1) init failed, (5) device not implemented

4.1.7 PAX X*

POSITION ABSOLUTE FOR X-AXIS:

Moves the X-axis to an absolute position, leaving the other axes positions unchanged.

X: dist. in 0.1mm [(-InitOffset + 10)...X-Range evaluated by ***ARX*** command]

EXAMPLE: #W2PAX1000

Absolute move in X-direction to 100mm from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.8 PAY Y*

POSITION ABSOLUTE FOR Y-AXIS:

Moves the Y-axis to an absolute position, leaving the other axes positions unchanged.

Y: dist. in 0.1mm [(-InitOffset + 10)...Y-Range evaluated by ***ARY*** command]

EXAMPLE: #W2PAY500

Absolute move in Y-direction with 50 mm from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.9 PAZ Z*

POSITION ABSOLUTE FOR Z-AXIS:

Moves the Z-axis to an absolute position, leaving the other axes positions unchanged. The zero position is somewhere near the worktable. CAUTION: use ARZ command for range evaluation before using a PAZ command.

Z: dist. in 0.1mm [- SubmergeRange …(Range – SubmergeRange)]

(SubmergeRange set by the ***SCS*** command)

EXAMPLE: #W2PAZ100

Absolute move in Z-direction to 10 mm above worktable (max. Z-range).

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.10 PAR R*

POSITION ABSOLUTE FOR R-AXIS:

Moves the Rotator-axis to an absolute position, leaving the other axes positions unchanged.

R: dist. in 0.1dg [(-7200...+7200]

EXAMPLE: #W2PAR500

Absolute move with Rotator-axis 50 dg from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.11 PAG G*

POSITION ABSOLUTE FOR G-AXIS:

Moves the Gripper-axis to an absolute position, leaving the other axes position unchanged.

G: dist. in 0.1mm [MinSpace..MaxSpace]

EXAMPLE: #W2PAG120

Gripper opens about 120mm

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.12 PRX +/- X*

POSITION RELATIVE FOR X-AXIS:

Moves X-axis relative to its actual position. All other positions remain unchanged.

+/-X: dist. in 0.1mm [within Range (-InitOffset + 10)...X-Range-InitOffset]

EXAMPLE: #W2PRX500

Move the X-axe 50 mm from the current position to the right.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.13 PRY +/- Y*

POSITION RELATIVE FOR Y-AXIS:

Moves Y- axis relative to its actual position. All other positions remain unchanged.

+/-Y: dist. in 0.1mm [within Range (-InitOffset + 10)...Y-Range-InitOffset]

EXAMPLE: #W2PRY200

Move the Y-axis 20 mm from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.14 PRZ +/- Z*

POSITION RELATIVE FOR Z-AXIS:

Move the Z-axis relative to its actual position, leaving the other axes unchanged. The zero position is somewhere near the worktable.

+/-Z: dist. in 0.1mm [within Range -SubmergeRange …(Range – SubmergeRange)]

EXAMPLE: #A1PRZ-100

Move the Z-axis down 10 mm from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.15 PRR +/- R*

POSITION RELATIVE FOR ROTATOR-AXIS:

Moves the Rotator-axis relative to its actual position. All other positions remain unchanged.

+/-R: dist. in 0.1dg [within Range (-7200 +7200)]

EXAMPLE: #W2PRR200

Move the Rotator-axis 20 dg from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.16 PRG +/- G*

POSITION RELATIVE FOR GRIPPER-AXIS:

Moves the Gripper-axis relative to its actual position. All other positions remain unchanged. Moving the gripper axis with relative commands sometimes creates rounding errors. However, the position reported by the ***RPG*** command always correlates with the actual gripper position.

+/-G: dist. in 0.1mm [within Range MinSpace..MaxSpace] set by the ***SGP*** Cmd

EXAMPLE: #W2PRG200

Move the Gripper-axis 20 mm from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.17 MAX X*, SlowSpeed*

MOVE ABSOLUTE SLOW SPEED FOR X-AXIS:

Moves the X-axis with a slow speed to an absolute position, leaving the other axes position unchanged. If no speed is selected, the default speed stored in memory is used.

X: dist. in 0.1mm [(-InitOffset + 10)...X-Range evaluated by ***ARX*** command]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 750)

EXAMPLE: #W2MAX1000,500

Absolute move in X-direction to 100mm from init position with a slow speed of 50mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.18 MAY Y*, SlowSpeed*

MOVE ABSOLUTE SLOW SPEED FOR Y-AXIS:

Moves the Y-axis with a slow speed to an absolute position, leaving the other axis position unchanged. If no speed is selected, the default speed stored in memory is used.

Y: dist. in 0.1mm [(-InitOffset + 10)...Y-Range evaluated by ***ARY*** command]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 1424)

EXAMPLE: #W2MAY1000,250

Absolute move in Y-direction to 100 mm from init position with a slow speed of 25 mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.19 MAZ Z*, SlowSpeed*

MOVE ABSOLUTE SLOW SPEED FOR Z-AXIS:

Moves the Z-axis with a slow speed to an absolute position, leaving the other axes position unchanged. If no speed is selected, the default speed stored in memory is used. The zero position is somewhere near the worktable.

Z: dist. in 0.1mm [- SubmergeRange …(Range – SubmergeRange)]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 880)

EXAMPLE: #W2MAZ100,200

Move the Z-axis down to absolute position 10 mm with a speed of 20 mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.20 MAR R*, SlowSpeed*

MOVE ABSOLUTE SLOW SPEED FOR ROTATOR-AXIS:

Moves the Rotator-axis with a slow speed to an absolute position, leaving the other axis position unchanged. If no speed is selected, the default speed stored in memory is used.

R: dist. in 0.1dg [-7200 +7200]

SlowSpeed: speed in 0.1dg/s [5...10000] (default = 780)

EXAMPLE: #W2MAR1000,250

Absolute move with Rotator axis to 100 mm from init position with a slow speed of 25 dg/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.21 MAG G*, SlowSpeed*

MOVE ABSOLUTE SLOW SPEED FOR GRIPPER-AXIS:

Moves the Gripper-axis with a slow speed to an absolute position, leaving the other axis position unchanged. If no speed is selected, the default speed stored in memory is used.

G: dist. in 0.1mm [MinSpace..MaxSpace]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 780)

EXAMPLE: #W2MAG1000,250

Absolute move with Gripper axis to 100 mm from init position with a slow speed of 25 mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.22 MRX +/- X*, SlowSpeed*

MOVE RELATIVE SLOW SPEED FOR X-AXIS:

Moves the X-axis with slow speed relative to its actual position, leaving the other axes unchanged. If no speed is selected, the default speed stored in memory is used.

+/-X: dist. in 0.1mm [within Range (-InitOffset + 10)...X-Range-InitOffset]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 750)

EXAMPLE: #W2MRX1000,500

Relative move in X-direction, 100mm from current position with a slow speed of 50mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.23 MRY +/- Y*, SlowSpeed*

MOVE RELATIVE SLOW SPEED FOR Y-AXIS:

Moves the Y-axis with slow speed relative to its actual position, leaving the other axes unchanged. If no speed is selected, the default speed stored in memory is used.

+/-Y: dist. in 0.1mm [within Range (-InitOffset + 10)...Y-Range-InitOffset]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 1424)

EXAMPLE: #W2MRY-500,50

Move Y-axis -50 mm from the current position back with a speed of 5 mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.24 MRZ +/- Z*, SlowSpeed*

MOVE RELATIVE SLOW SPEED FOR Z-AXIS:

Moves the Z-Axis to positions relative to their present position, leaving the other axes unchanged. If no speed is selected, the default speed stored in memory is used. The zero position is somewhere near the worktable.

+/-Z: dist. in 0.1mm [within Range -SubmergeRange …(Range – SubmergeRange)]

SlowSpeed: speed in 0.1mm/s [5...10000] (default = 880)

EXAMPLE: #W2MRZ100,200

Move Z-axis up from the current position with a speed of 20 mm/s.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.25 MRR +/- R*

MOVE RELATIVE SLOW SPEED FOR R-AXIS:

Moves the Rotator-axis with slow speed relative to its actual position, leaving the other axes unchanged. If no speed is selected, the default speed stored in memory is used.

+/-R: dist. in 0.1dg [within Range (-7200 +7200)]

SlowSpeed: speed in 0.1dg/s [5...10000] (default = 780)

EXAMPLE: #W2MRR-500

Move Rotator-axe -50 mm from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.26 MRG +/- G*,

MOVE RELATIVE SLOW SPEED FOR G-AXIS:

Moves the Gripper-axis with slow speed relative to its actual position, leaving the other axes unchanged. If no speed is selected, the default speed stored in memory is used.

+/-G: dist. in 0.1mm [within Range MinSpace..MaxSpace] set by the ***SGP*** Cmd

SlowSpeed: speed in 0.1dg/s [5...10000] (default = 780)

EXAMPLE: #W2MRG-500

Move Gripper-axis -50 mm from the current position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init

4.1.27 PAA X*,Y*,Z*,R*,G*

POSITIONING ALL AXIS SIMULTANEOUSLY ABSOLUTE:

Moves all axis with normal speed to an absolute position. If aparameter is not defined, the related keeps its current position unchanged. However, rotator and gripper movements are not allowed simultaneously. CAUTION: Incorrect usage of this command may damage the instrument.

X: dist. in 0.1mm [- X-Range ... + X-Range set by ***ARX*** cmd]

Y: dist. in 0.1mm [- Y-Range ... +Y-Range set by ***ARY*** cmd]

Z: dist. in 0.1mm [-SubmergeRange …(Z-Range – SubmergeRange)]

R: dist. in 0.1deg [- 7200..7200]

G: dist. in 0.1mm [MinSpace..MaxSpace]

EXAMPLE: #W2PAA2000, 3000,1000,,250

Moves all axis to the selected absolute position. The rotator axis remains untouched.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init, (18) invalid axis combination

4.1.28 PRA X*,Y*,Z*,R*,G*

POSITIONING ALL AXIS SIMULTANEOUSLY RELATIVE:

Moves all axis with normal speed to a relative position. If aparameter is not defined, the related axis keeps its current position unchanged. However, rotator and gripper movements are not allowed simultaneously. CAUTION: Incorrect usage of this command may damage the instrument.

X: dist. in 0.1mm [within Range - X-Range ... + X-Range set by ***ARX*** cmd]

Y: dist. in 0.1mm [within Range - Y-Range ... +Y-Range set by ***ARY*** cmd]

Z: dist. in 0.1mm [within Range -SubmergeRange …(Z-Range – SubmergeRange)]

R: dist. in 0.1deg [within Range - 7200..7200]

G: dist. in 0.1mm [within Range MinSpace..MaxSpace]

EXAMPLE: #W2PAA-1000, 800,-1000,-900,

Moves all axis for the selected relative position. The gripper axis stays untouched.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init, (18) invalid axis combination

4.1.29 MAA X*,Y*,Z*,R*,G*

MOVE ALL AXIS SIMULTANEOUSLY SLOW ABSOLUTE:

Moves all axis with slow speed to an absolute position. If aparameter is not defined, the related keeps its current position unchanged. However, rotator and gripper movements are not allowed simultaneously. This command can be used to move centrifuge buckets. CAUTION: Incorrect usage of this command may damage the instrument.

X: dist. in 0.1mm [- X-Range ... + X-Range set by ***ARX*** cmd]

Y: dist. in 0.1mm [- Y-Range ... +Y-Range set by ***ARY*** cmd]

Z: dist. in 0.1mm [-SubmergeRange …(Z-Range – SubmergeRange)]

R: dist. in 0.1deg [- 7200..7200]

G: dist. in 0.1mm [MinSpace..MaxSpace]

EXAMPLE: #W2PAA2000, 3000,1000,,250

Moves all axis to the selected absolute position. The rotator axis stays untouched.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init, (18) invalid axis combination

4.1.30 MRA X*,Y*,Z*,R*,G*

MOVE ALL AXIS SIMULTANEOUSLY SLOW RELATIVE:

Moves all axis with slow speed to a relative position. If aparameter is not defined, the related axis keeps its current position unchanged. However, rotator and gripper movements are not allowed simultaneously. This command can be used to move centrifuge buckets. CAUTION: Incorrect usage of this command may damage the instrument.

X: dist. in 0.1mm [within Range - X-Range ... + X-Range set by ***ARX*** cmd]

Y: dist. in 0.1mm [within Range - Y-Range ... +Y-Range set by ***ARY*** cmd]

Z: dist. in 0.1mm [within Range -SubmergeRange …(Z-Range – SubmergeRange)]

R: dist. in 0.1deg [within Range - 7200..7200]

G: dist. in 0.1mm [within Range MinSpace..MaxSpace]

EXAMPLE: #W2PAA-1000, 800,-1000,-900,

Moves all axis for the selected relative position. The gripper axis stays untouched.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand, (7) device not init, (18) invalid axis combination

4.1.31 AGR GripSearch*, GripOpen*

GRIPP TUBE:

This command grips with the gripper drive at the current X, Y, Z position. The gripper searches for a tube to the absolute GripSearch position. If there is no tube found, it re-opens to the absolute GripOpen position. An error message (tube not fetched) is sent back.

GripSearch: [MinSpace..MaxSpace]

GripOpen: [MinSpace..MaxSpace]

EXAMPLE: #W2AGR

Grips a tube at the current position.

GENERATED ERRORS: (3) invalid operand, (7) device not init, (17) tube not fetched

4.1.32 APC GripSearch*,GripOpen*,Zpick*,Zup*,Rotation*, RotComp*

PICK TUBE:

This command grips with the gripper drive at the current X, Y position. The gripper moves down to the absolute *Zpick* with a Gripper spacing defined by the absolute *GripOpen* parameter. Then it searches for a tube until the *GripSearch* position is achieved. If no tube is found an error message is sent back and the gripper re-opens to the defined *GripOpen* position. The Z-axis stays at *Zpick*. If a tube is gripped successfully the z-axis moves up to for a tube to the absolute *Zup* position and the *Rotation* is performed. The *RotComp* parameter modifies the InitOffset of the R-axis while a tube is gripped! The *InitOffset* is set back to its original value after a ***APL*** command or after a ***PIA*** or ***PIR*** command takes place.

GripOpen: Gripper Spacing in 0.1mm [MinSpace…MaxSpace]

GripSearch: Gripper Spacing in 0.1m [MinSpace…MaxSpace]

Zpick: Pick hight in 0.1mm above worktable

Zup: Z Hight in 0.1mm after a successful pick

Rotation: Rotation absolute in 0.1deg before pick [-7200..+7200]

RotComp: Compensate rotation while grip in 0.1 deg[0… 300]

EXAMPLE: #W2APC300,120,1000,2500,1800,50

GENERATED ERRORS: (3) invalid operand, (7) device not init, (17) tube not

4.1.33 APL GripOpen*, Zplace*, Zup*, Rotation*, TubCheck*, WipeOffDist*, TubeLeftChck*

PLACE TUBE:

This command places a tube at the current X, Y position. The gripper moves down to the absolute *Zplace*. While it spaces to the *GripOpen* position, the tube is released. It again grips and releases the tube if the *TubCheck* parameter is set. An error message (tube not fetched) is sent back if no tube is found. The gripper re-opens to the absolute GripOpen position while the Z position stays at *Zplace*. If the WipeOff parameter is set Z moves down in order to wipe off a sticky tube. The Z-axis moves up to the absolute Zup position if the tube check is successful or *TubCheck* parameter is not set.

GripOpen: Gripper Spacing in 0.1mm [MinSpace...*MaxSpace*]

Zplace: Place hight in 0.1mm above worktable

Zup: Z Hight in 0.1mm after a successful place

Rotation: Rotation absolute in 0.1deg before place [-7200..+7200]

TubeCheck: [0..1] 0=inactive, 1=active [0,1]

WipeOffDist: [-100..0]

TubeLeftChck:[0..1] 0=inactive, 1=active [0,1]

EXAMPLE: #W2APC270,120,1000,2500,1800,1,-20,1

Place a tube at Z-position 300 and a rotation of 250 degree, check if the tube is still in the gripper, Open gripper 27mm, move 2mm down in order to wipe off sticky tubes, move to Z-position 1000, close gripper to check for left tube and open it again for 270mm.

GENERATED ERRORS: (3) invalid operand, (7) device not init, fetched, (16) Tube left in gripper (17) tube not fetched

4.1.34 ATD Diameter

TEACH DIAMETER:

This command teaches the gripper to a known tube diameter at the current X, Y, Z position. The gripper searches for a tube. This command should be used after a collision or exchange of a gripper tool. The calculated factor is used only for the ***RTD*** command but not for gripper positioning! The diameter for teaching needs to be set to 13.0mm.

Diameter: 130 [in 0.1mm]

EXAMPLE: #W2ATD130
Grips a tube at the current position.

GENERATED ERRORS: (3) invalid operand, (7) device not init, (17) tube not fetched

4.1.35 ARX

AUTO RANGE X-AXIS:

This command moves to the right and the left stopper of the X-axis and sets the X-range. The value can be stored into the non-volatile memory with the ***AWE*** command.

EXAMPLE: #W2ARX
Auto ranges X-axis.

GENERATED ERRORS: none

4.1.36 ARY

AUTO RANGE Y-AXIS:

This command moves to the right and the left stopper of the Y-axis and sets the the Y-range. The value can be stored into the non-volatile memory with the ***AWE*** command.

EXAMPLE: #W2ARY
Auto ranges X-axis.

GENERATED ERRORS: none

4.1.37 ARZ

AUTO RANGE Z-AXIS:

This gripper moves down to the worktable, touches the surface and moves up to the upper stopper. The worktable surface is now 0.

The value can be stored into the non-volatile memory with the ***AWE*** command.

EXAMPLE: #W2ARZ
Auto ranges Z-axis.

GENERATED ERRORS: none

4.2 Set Commands

4.2.1 SGP PWMLimit*, GripSpeed*, MinSpace*, MaxSpace*

SET GRIP PARAMETER:

The *SGP* command sets the PWM limit, the gripper speed, the minimum gripper spacing and the maximum gripper spacing in the volatile memory. The parameters can be stored in non-volatile memory by the ***AWE*** command. CAUTION: incorrect gripper parameter settings might damage the servo drives!

PWMLimit:* PWM Limit [0...249] (default = 80)

*GripSpeed *:* GripperSpeed [50..5000] (default= 3000)

*MinSpace *:* Minimum gripper spacing (default=70)

*MaxSpace *:* Maximum gripper spacing (default=280)

EXAMPLE: #W2SGP80, 3000, 90, 280

GENERATED ERRORS: (3) invalid operand

4.2.2 SFX EndSp*, Accel*

SET X-RAMP PARAMETERS:

This command sets the X-axis DC-motor ramp settings in volatile memory.

EndSp:* speed in 0.1mm/s. [50....16000] (default = 15000)

Accel:* acceleration in 0.1mm/ s². [50..16000] (default = 1500)

EXAMPLE: #W2SFX5000,1000
Set End Speed to 500mm/s and the acceleration to 100mm/s².

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.3 SFY EndSp*, Accel*

SET Y-RAMP PARAMETERS:

This command sets the Y-axis DC-motor ramp settings in volatile memory.

EndSp:* speed in 0.1mm/s. [50....10000] (default = 7449)

Accel:* acceleration in 0.1mm/s². [50..10000] (default = 5385)

EXAMPLE: #W2SFY1000,500
Set End Speed to 100mm/s and the acceleration to 50mm/s².

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.4 SFZ EndSp*, Accel*

SET Z-RAMP PARAMETERS:

This command sets the Z-axis DC-motor ramp settings in volatile memory.

EndSp:* speed in 0.1mm/s. [50....10000] (default = 6614)

Accel:* acceleration in 0.1mm/s². [50..10000] (default = 6614)

EXAMPLE: #W2SFZ2500,3000

Set end Speed to 250mm/s and the acceleration to 300mm/s²..

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.5 SFR EndSp*, Accel*

SET ROTATOR-RAMP PARAMETERS:

This command sets the R-axis DC-motor ramp settings in volatile memory. This speed is also use by the gripper since gripper and rotator needs the same speed and acceleration for its dependent movements.

EndSp:* speed in 0.1dg/s. [50....10000] (default = 7798)

Accel:* acceleration in 0.1dg/ s².. [50..10000] (default = 7798)

EXAMPLE: #W2SFR2500,3000
Set end speed to 250mm/s and the acceleration to 300dg/s².

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.6 SOX InitOffset

SET INITIALIZATION OFFSET FOR X-AXIS:

This command sets the initialization offset for the X-axis. This parameter can be stored in non volatile memory by the ***AWE*** command.

InitOffset: dist. in 0.1mm [0...1200] (default = 100)

EXAMPLE: #W2SOX100
Set the X-Init offset to 10 mm from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.7 SOY InitOffset

SET INITIALIZATION OFFSET FOR Y-AXIS:

This command sets the initialization offset for the Y-axis. This parameter can be stored in non volatile memory by the ***AWE*** command.

InitOffset: dist. in 0.1mm [0...1200] (default = 100)

EXAMPLE: #W2SOY200
Set the Y-Init offset to 20 mm from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.8 SOZ InitOffset

SET INITIALIZATION OFFSET FOR Z-AXIS:

This command sets the initialization offset for the Z-axis. This parameter can be stored in non volatile memory by the ***AWE*** command.

InitOffset: dist. in 0.1mm [0...1200] (default = 50)

EXAMPLE: #W2SOZ100
Set the Z-Init offset to 10 mm from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.9 SOR InitOffset

SET INITIALIZATION OFFSET FOR ROTATOR-AXIS:

This command sets the initialization offset for the R-axis. This parameter can be stored in non volatile memory by the ***AWE*** command. The ***SOR*** command must not be used while a tube is gripped with the ***APC*** command!

InitOffset: dist. in 0.1dg [0...1200] (default = 100)

EXAMPLE: #W2SOR100
Set the R-Init offset to 10 dg from init position.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.10 SAX ScaleAdjust

SET SCALE ADJUST FACTOR FOR X-AXIS
This command is used for the X-axis scale factor fine-tuning. This value can be stored into the non-volatile memory with the ***AWE*** command.

ScaleAdjust: [10000 +/- 5%] (default = 10000)

EXAMPLE: #W2SAX9975
Set the scale adjust factor to 0,9975.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.11 SAY ScaleAdjust

SET SCALE ADJUST FACTOR FOR Y-AXIS
This command is used for the Y-axis scale factor fine-tuning. This value can be stored into the non-volatile memory with the ***AWE*** command.

ScaleAdjust: *[*10000 +/- 5%] (default = 10000)

EXAMPLE: #W2SAY10055
Set the scale adjust factor to 1,0055.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.12 SXD Displacement

SET DISPLACEMENT FOR X AXIS
The X-displacement will be subtracted from the x-coordinate of any absolute X-movement or X-

positioning command. This allows to operate different P&P arms on the same X-axis using a common X-zero position. CAUTION: incorrect displacement parameter settings might damage the instrument! The value can be stored into the non-volatile memory with the ***AWE*** command.

Displacement: [0..20000] (default = 0)

EXAMPLE: #W2SXD600
Set the x-displacemt 600 mm to the right.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.13 SMX Range

SET OVERALL MACHINE LIMITATION X-AXIS ONLY
The ***SMX*** command sets the physical limitation range (in the volatile memory), for the X-axis to ensure the X-axis range can not be set to invalid machine positions. After initial firmware download, the ***SMX*** range is set to the default value. After definition, the maximal range can be stored into the non-volatile memory with the ***AWE*** command. See also ***ARX*** for automated range setting.

Range: distance in 0.1 mm [0**..**100'000] (default = 15000)

EXAMPLE: #W2SMX10000
Set machine limitation in X-axis to 1000 mm.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.14 SMY Range

SET OVERALL MACHINE LIMITATION Y-AXIS ONLY
The ***SMY*** command sets the physical limitation range (in the volatile memory), for the Y-axis of the P&P arm to ensure the Y-axis range can not be set to invalid machine positions. After initial firmware download, the ***SMY*** range is set to the default value. After definition, the maximal range can be stored into the non-volatile memory with the ***AWE*** command. See also ***ARY*** for automated range setting.

Range: distance in 0.1 mm [0**..**100'000] (default = 10000)
EXAMPLE: #W2SMY10000
Set machine limitation in Y-axis to 1000 mm.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.15 SMZ Range

SET OVERALL MACHINE LIMITATION Z-AXIS ONLY
The ***SMZ***command sets the physical limitation range (in the volatile memory), for the Z-axis of the P&P arm to ensure the Z-axis range can not be set to invalid machine positions. Submerge range subtracted from the Z-range gives the Z-range above the worktable. After initial firmware download, the ***SMY*** range is set to the default value. After definition, the maximal range can be stored into the non-volatile memory with the ***AWE*** command. See also ***ARZ*** for automated range setting.

Range: distance in 0.1 mm [0..100'000] (default = 3800)
EXAMPLE: #W2SMZ3800
Set machine limitation in Z-axis to 3800 mm.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.16 SCO AxisNo*

SET CURRENT OFF:

This command sets the chosen axis off power. If no parameter is entered all axis or power off.

AxisNo: 1 = X-axis

2 = Y-axis

3 = Z-axis

4 = R-axis

5 = G-axis

EXAMPLE: #W2SCO
All axis power off.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.17 SCS SubmergeRange

SET CENTRIFUGE SUBMERGE RANGE
The SCS command sets the physical limitation range (in the volatile memory), for the Z-axis submerge. (Steps below the worktable surface). After definition, the maximal range can be stored into the non-volatile memory with the ***AWE*** command.

SubmergeRange: submerge range in 0.1 mm [1000..1400] (default = 1200)

EXAMPLE: #W2SCS1200
Set submerge range to 120 mm.

RESPONSE: none

GENERATED ERRORS: (3) invalid operand

4.2.18 SSO OnOff

SET SOLENOID ON/OFF
The SSO command sets solenoid output on or off. The solenoid controls the Z-break.

OnOff: 0,1 (0=off;1=on)

EXAMPLE: #W2SSO1
Release Z-break

RESPONSE: none
GENERATED ERRORS: (3) invalid operand

4.3 REPORT COMMANDS

4.3.1 RFV Selector

REPORT FIRMWARE VERSION

This command reports the Firmware and Bootware Version. Also the instruments parameters can be reported.

Selector: 0 = Firmware Version (default)
1 = Bootware Version
2 = Serial Number
3 = Hardware Version
4 = TypeId
5 = Copyright
6 = Address Switch

EXAMPLE: #W2RFV0
Reports the actual firmware version.

RESPONSE: "PPU-Vx.xx-MM/YY" for Pick&Place Unload
“PPL_Vx.xx-MM/YY” for Pick&Place Load

GENERATED ERRORS: (3) invalid operand

4.3.2 RDF Selector

REPORT DIAGNOSTIC FUNCTIONS:

All the diagnostic values will be saved at power off of the instrument or if a *Power fail* occurs. The stored data can be read by this command.

Selector: 0 = Number of power ups (default)
1 = Report hour meter.

EXAMPLE: #P1RDF1
Reports the sum of *switched on* time in hours.

RESPONSE: number of hours

GENERATED ERRORS: (3) invalid operand

4.3.3 REE .

REPORT EXTENDED ERROR CODE
Get axis specific errors from a DC-Servo drive. To visualize the error codes as ASCII characters, an offset of 40Hex is added to the error value. See error table of the DC-Servo Command Set (TECAN DocNo. 390 249)

EXAMPLE: #W2REE
RESPONSE: "@@@@@" = all axis are initialized and OK
"@@G@@" = Z-axis is not initialized

GENERATED ERRORS: none

4.3.4 RPX Selector*

REPORT CURRENT PARAMETER FOR X-AXIS:

This command reports the current parameter of the X-axis. Only one axis parameter can be reported at a time.

Selector: 0 = Report current position in 1/10 mm

1 = Report acceleration in 1/10 mm/s2

2 = Report end speed in 1/10 mm/s

3 = Report init speed in 1/10 mm/s

4 = Report init offset in 1/10 mm

5 = Report actual machine range in 1/10 mm

6 = Report init error steps in encoder increments

7 = Report travel position in 1/10 mm

8 = Report scale factor

9 = Slow Speed

EXAMPLE: #W2RPX2
Reports the end speed of X-drive.

RESPONSE: one X-axis parameter.

GENERATED ERRORS: (3) invalid operand

4.3.5 RPY Selector*

REPORT CURRENT PARAMETER FOR Y-AXIS:

This command reports the current parameter of the Y-axis. Only one axis parameter can be reported at a time.

Selector: 0 = Report current position in 1/10 mm

1 = Report acceleration in 1/10 mm/s2

2 = Report end speed in 1/10 mm/s

3 = Report init speed in 1/10 mm/s

4 = Report init offset in 1/10 mm

5 = Report actual machine range in 1/10 mm

6 = Report init error steps in encoder increments

7 = Report travel position in 1/10 mm

8 = Report scale factor

9 = Slow Speed

EXAMPLE: #W2RPY2
Reports the end speed of Y-drive.

RESPONSE: one Y-axis parameter.

GENERATED ERRORS: (3) invalid operand

4.3.6 RPZ Selector*

REPORT CURRENT PARAMETER FOR Z-AXIS:

This command reports the current parameter of the Z-axis. Only one axis parameter can be reported at a time.

Selector: 0 = Report current position in 1/10 mm

1 = Report acceleration in 1/10 mm/s2

2 = Report end speed in 1/10 mm/s

3 = Report init speed in 1/10 mm/s

4 = Report init offset in 1/10 mm

5 = Report actual machine range in 1/10 mm

6 = Report init error steps in encoder increments

7 = Report travel position in 1/10 mm

8 = Report scale factor

9 = Slow Speed

EXAMPLE: #W2RPZ2
Reports the end speed of Z-drive.

RESPONSE: one Z-axis parameter.

GENERATED ERRORS: (3) invalid operand

4.3.7 RPR Selector*

REPORT CURRENT PARAMETER FOR ROTATOR-AXIS:

This command reports the current parameter of the Rotator-axis. Only one axis parameter can be reported at a time.

Selector: 0 = Report current position in 1/10 dg

1 = Report acceleration in 1/10 dg/s2

2 = Report end speed in 1/10 dg/s

3 = Report init speed in 1/10 dg/s

4 = Report init offset in 1/10 dg

5 = Report actual machine range in 1/10 dg

6 = Report init error steps in encoder increments

7 = Report travel position in 1/10 dg

8 = Report scale factor

9 = Slow Speed

EXAMPLE: #W2RPR2
Reports the end speed of rotator drive.

RESPONSE: one R-axis parameter.

GENERATED ERRORS: (3) invalid operand

4.3.8 RPG Selector*

REPORT CURRENT PARAMETER FOR GRIPPER-AXIS:

This command reports the current parameter of the gripper axis. Only one axis parameter can be reported at a time.

Selector: 0 = Report current position in 1/10 mm

1 = Report acceleration in 1/10 mm/s2

2 = Report end speed in 1/10 mm/s

3 = Report init speed in 1/10 mm/s

4 = Report init offset in 1/10 mm

5 = Report actual machine range in 1/10 mm

6 = Report init error steps in encoder increments

7 = Report travel position in 1/10 mm

8 = Report scale factor

9 = Slow Speed

EXAMPLE: #W2RPG2
Reports the end speed of gripper drive.

RESPONSE: one G-axis parameter.

GENERATED ERRORS: (3) invalid operand

4.3.9 RGP Selector

REPORT GLOBAL GRIPPER PARAMETER:

This command reports all Gripper Parameters

Selector: 0 = MinSpace

1 = MaxSpace

2 = GrippPWMLimit

3 = GripSpeed

4 = RotatOffset

2 = DiameterCal

6 = DiaTeachCorr

7 = GripperFlag

EXAMPLE: #W2RGP1
Report gripper MinSpace

RESPONSE: PWMLimit

GENERATED ERRORS: (3) invalid operand

4.3.10 RTD

REPORT TUBE DIAMETER

This command reports the diameter of a gripped tube. See ATD command for diameter calibration.

EXAMPLE: #W2RTD

RESPONSE: Diameter [0.1 mm].

GENERATED ERRORS: none

4.3.11 RXD

REPORT DISPLACEMENT FOR X AXIS
The X-displacement will be subtracted from the x-coordinate of any absolute X-movement or X-positioning command. It is set by the ***SXD*** command. This allows to operate different P&P arms on the same X-axis using a common X-zero position.

Displacement: [0..20000] (default = 0)

EXAMPLE: #W2RXD

RESPONSE: Displacement [0.1 mm].

GENERATED ERRORS: none

5 Error Codes and Messages

Error	Mnemonic	Description and Comments
1	Initialization Error	
2	Invalid Command	
3	Invalid Operand	
4	Invalid Cmd Sequence	
5	Device Not Implemented	
6	Timeout Error	
7	Device not initialized	
9	Break point not reached	
10	Drive no load	Drive with no load detected. <REE> reports which drive.
12	No access to parameter pages	No access to Flash parameter pages
15	Command overflow	An action command was sent while another action command is still pending
16	Tube Left	Tube left back in gripper after place command
17	Tube not fetched	The tube was not succesfully fetched within the specified gripper range.
18	Invalid axis combination	Rotator an Gripper movements are not allowed simultaneously.